

4.6 Vectors

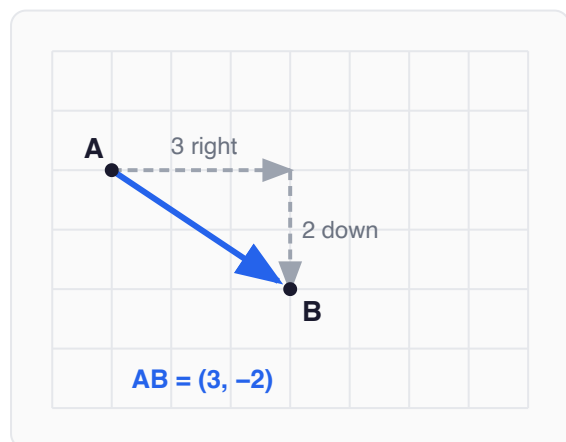
AQA GCSE Mathematics 8300 — Geometry & Measures — spec refs G24, G25 — Papers 1 (non-calculator), 2 and 3 (calculator)

4.6.1 What Is a Vector?

A **vector** is a quantity that has both **magnitude** (size) and **direction**. A **scalar** has size only — it is just a number, such as 3, -2 or $\frac{1}{2}$.

Vectors can be written in three ways:

- **Column vector:** $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ — the top number is the movement in the x -direction (right is positive, left is negative) and the bottom number is the movement in the y -direction (up is positive, down is negative).
- **Bold lower-case letter:** **a**. In handwriting you cannot write in bold, so **underline** it: a.
- **Between two points:** $\overrightarrow{AB}$ means the vector that starts at A and finishes at B .



Column vector from $A(x_1, y_1)$ to $B(x_2, y_2)$:

$$\overrightarrow{AB} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$$

Reversing a vector: $\overrightarrow{BA} = -\overrightarrow{AB}$

Worked Example: A is the point $(2, 1)$ and B is the point $(5, -3)$. Write down $\overrightarrow{AB}$ and $\overrightarrow{BA}$.

$$\overrightarrow{AB} = \begin{pmatrix} 5 - 2 \\ -3 - 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \end{pmatrix} \quad \text{and} \quad \overrightarrow{BA} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$$

Equal and Negative Vectors

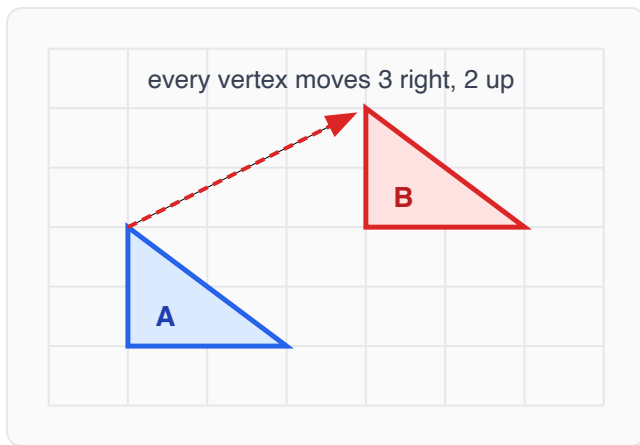
Two vectors are **equal** if they have the same magnitude *and* the same direction — they do not have to be in the same place. In a parallelogram $ABCD$, $\overrightarrow{AB} = \overrightarrow{DC}$ even though they are opposite sides.

The vector $-\mathbf{a}$ has the same length as $\mathbf{a}$ but points the **opposite way**. If $\mathbf{a} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ then $-\mathbf{a} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$.

4.6.2 Translations as Vectors (G24)

A **translation** slides every point of a shape the same distance in the same direction. It is described fully by the word **translation** and a **column vector**. The image is congruent to the object and has the same orientation.

Translating the point (x, y) by the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ gives the image $(x + a, y + b)$.



Triangle A is mapped onto triangle B by the **translation** $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$. To go back from B to A use $\begin{pmatrix} -3 \\ -2 \end{pmatrix}$.

Worked Example: A triangle has vertices $(1, 1)$, $(3, 1)$ and $(1, 4)$. It is translated by $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$.

Find the vertices of the image.

Add -4 to every x -coordinate and 2 to every y -coordinate:

$$(1, 1) \rightarrow (-3, 3), \quad (3, 1) \rightarrow (-1, 3), \quad (1, 4) \rightarrow (-3, 6)$$

Worked Example: Shape P is translated by $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$ and then by $\begin{pmatrix} -5 \\ 1 \end{pmatrix}$. Find the single translation that has the same effect.

Add the vectors: $\begin{pmatrix} 2 \\ -3 \end{pmatrix} + \begin{pmatrix} -5 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \end{pmatrix}$

Exam Tip: "Describe fully" a translation needs two things: the word **translation** and the **column vector**. Do not write it as a coordinate like $(3, 2)$ and never use words like "moved" or "shifted" instead of "translation". Count squares carefully from one vertex to its image, not from the edge of one shape to the edge of the other.

4.6.3 Adding, Subtracting and Multiplying by a Scalar (G25)

Column Vectors

Addition: $\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a + c \\ b + d \end{pmatrix}$ **Subtraction:** $\begin{pmatrix} a \\ b \end{pmatrix} - \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a - c \\ b - d \end{pmatrix}$

Multiplying by a scalar k : $k \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$

Worked Example: $\mathbf{a} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$. Work out $2\mathbf{a} - 3\mathbf{b}$.

$$2\mathbf{a} = \begin{pmatrix} 4 \\ 10 \end{pmatrix}, \quad 3\mathbf{b} = \begin{pmatrix} -3 \\ 9 \end{pmatrix}$$

$$2\mathbf{a} - 3\mathbf{b} = \begin{pmatrix} 4 - (-3) \\ 10 - 9 \end{pmatrix} = \begin{pmatrix} 7 \\ 1 \end{pmatrix}$$

Worked Example: $\mathbf{p} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$. Find $\mathbf{r}$ such that $\mathbf{p} + 2\mathbf{r} = \mathbf{q}$.

Rearrange like an equation: $2\mathbf{r} = \mathbf{q} - \mathbf{p} = \begin{pmatrix} -8 \\ 3 \end{pmatrix}$, so $\mathbf{r} = \begin{pmatrix} -4 \\ 1.5 \end{pmatrix}$

Worked Example: Find m and n such that $m \begin{pmatrix} 2 \\ -1 \end{pmatrix} + n \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 3 \end{pmatrix}$.

Top: $2m + n = 8$. Bottom: $-m + 3n = 3$, so $m = 3n - 3$.

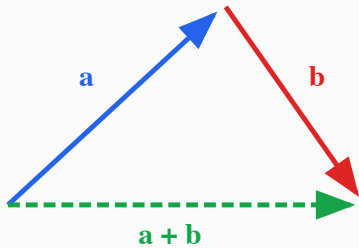
Substitute: $2(3n - 3) + n = 8 \Rightarrow 7n = 14 \Rightarrow n = 2$, then $m = 3$.

Check: $3 \begin{pmatrix} 2 \\ -1 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 3 \end{pmatrix} \checkmark$

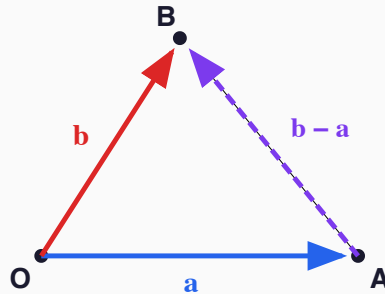
Vectors on Diagrams: the Triangle Law

Adding two vectors means following one and then the other. The **resultant** $\mathbf{a} + \mathbf{b}$ goes straight from the start of $\mathbf{a}$ to the end of $\mathbf{b}$. Subtracting $\mathbf{b}$ means adding $-\mathbf{b}$ (travelling backwards along $\mathbf{b}$).

Triangle law



From A to B via O



If $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$ then

$$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = -\mathbf{a} + \mathbf{b} = \mathbf{b} - \mathbf{a}$$

"End minus start" — exactly like the column vector between two points.

Worked Example: $OABC$ is a parallelogram with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$. Find $\overrightarrow{OB}$, $\overrightarrow{AC}$ and $\overrightarrow{CB}$.

Opposite sides of a parallelogram are equal vectors, so $\overrightarrow{AB} = \mathbf{c}$ and $\overrightarrow{CB} = \mathbf{a}$.

$$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = \mathbf{a} + \mathbf{c}$$

$$\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC} = -\mathbf{a} + \mathbf{c} = \mathbf{c} - \mathbf{a}$$

Worked Example: $ABCD$ is a parallelogram with $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AD} = \mathbf{d}$. E is the midpoint of CD . Find $\overrightarrow{BE}$.

$$\overrightarrow{DC} = \mathbf{a}, \text{ so } \overrightarrow{DE} = \frac{1}{2}\mathbf{a} \text{ and } \overrightarrow{AE} = \mathbf{d} + \frac{1}{2}\mathbf{a}.$$

$$\overrightarrow{BE} = \overrightarrow{BA} + \overrightarrow{AE} = -\mathbf{a} + \mathbf{d} + \frac{1}{2}\mathbf{a} = \mathbf{d} - \frac{1}{2}\mathbf{a}$$

Vector expressions simplify just like algebra: collect like terms and expand brackets.

Worked Example: Simplify $3(\mathbf{a} + 2\mathbf{b}) - 2(\mathbf{a} - \mathbf{b})$.

$$= 3\mathbf{a} + 6\mathbf{b} - 2\mathbf{a} + 2\mathbf{b} = \mathbf{a} + 8\mathbf{b}$$

Exam Tip: To find any vector on a diagram, pick a **route** along vectors you know. Going **with** an arrow uses $+$, going **against** an arrow uses $-$. Write the route first (e.g. $\overrightarrow{BE} = \overrightarrow{BA} + \overrightarrow{AE}$) — this earns method marks even if you slip later.

4.6.4 Parallel Vectors and Collinear Points

Higher

Two vectors are **parallel** if one is a **scalar multiple** of the other. For example $6\mathbf{a} + 3\mathbf{b} = 3(2\mathbf{a} + \mathbf{b})$, so $6\mathbf{a} + 3\mathbf{b}$ is parallel to $2\mathbf{a} + \mathbf{b}$ and three times as long. A negative multiple means parallel but in the **opposite direction**.

Points are **collinear** if they lie on the same straight line. To prove A , B and C are collinear:

1. Find two vectors joining the points, e.g. $\overrightarrow{AB}$ and $\overrightarrow{BC}$.
2. Show one is a multiple of the other (so they are parallel).
3. State that they **share a common point** (B), so the points lie on one line.

Worked Example: $\overrightarrow{AB} = 3\mathbf{a} + 2\mathbf{b}$ and $\overrightarrow{BC} = 6\mathbf{a} + 4\mathbf{b}$. Prove that A , B and C are collinear.

$$\overrightarrow{BC} = 2(3\mathbf{a} + 2\mathbf{b}) = 2\overrightarrow{AB}, \text{ so } \overrightarrow{AB} \text{ is parallel to } \overrightarrow{BC}.$$

They share the point B , so A , B and C are collinear. (Also $AB : BC = 1 : 2$.)

Worked Example: For what value of k is $2\mathbf{a} + 3\mathbf{b}$ parallel to $k\mathbf{a} + 12\mathbf{b}$?

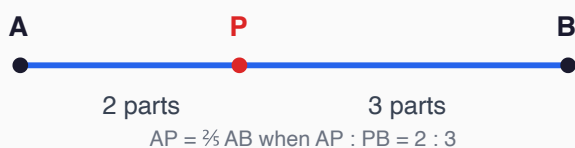
$$12\mathbf{b} = 4 \times 3\mathbf{b}, \text{ so the multiplier is } 4 \text{ and } k = 4 \times 2 = 8.$$

4.6.5 Midpoints and Ratios Along a Line Higher

If P lies on AB with $AP : PB = m : n$, then

$$\overrightarrow{AP} = \frac{m}{m+n} \overrightarrow{AB}$$

For the midpoint M of AB : $\overrightarrow{AM} = \frac{1}{2} \overrightarrow{AB}$, and with $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$: $\overrightarrow{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$.



Worked Example: $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and P is on AB with $AP : PB = 2 : 3$. Find $\overrightarrow{OP}$.

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} \text{ and } \overrightarrow{AP} = \frac{2}{5}(\mathbf{b} - \mathbf{a})$$

$$\overrightarrow{OP} = \mathbf{a} + \frac{2}{5}\mathbf{b} - \frac{2}{5}\mathbf{a} = \frac{3}{5}\mathbf{a} + \frac{2}{5}\mathbf{b}$$

Exam Tip: A ratio $m : n$ splits the line into $m + n$ parts. The most common error is using $\frac{m}{n}$ instead of $\frac{m}{m+n}$ — for $AP : PB = 1 : 3$, P is $\frac{1}{4}$ of the way along, not $\frac{1}{3}$.

4.6.6 Vector Proofs Higher

Vector proof questions usually ask you to show that two lines are **parallel**, that points are **collinear**, or to find a **ratio**. The method is always the same: express every relevant vector in terms of the given vectors, simplify, then look for a scalar multiple.

Worked Example: OAB is a triangle with $\overrightarrow{OA} = 3\mathbf{a}$ and $\overrightarrow{OB} = 3\mathbf{b}$. P is on OA with $OP : PA = 1 : 2$ and Q is on AB with $AQ : QB = 2 : 1$. Prove that PQ is parallel to OB .

$$\overrightarrow{PA} = \frac{2}{3} \times 3\mathbf{a} = 2\mathbf{a}$$

$$\overrightarrow{AB} = 3\mathbf{b} - 3\mathbf{a}, \text{ so } \overrightarrow{AQ} = \frac{2}{3}(3\mathbf{b} - 3\mathbf{a}) = 2\mathbf{b} - 2\mathbf{a}$$

$$\overrightarrow{PQ} = \overrightarrow{PA} + \overrightarrow{AQ} = 2\mathbf{a} + 2\mathbf{b} - 2\mathbf{a} = 2\mathbf{b}$$

$$\overrightarrow{OB} = 3\mathbf{b}, \text{ so } \overrightarrow{PQ} = \frac{2}{3}\overrightarrow{OB}: \text{ a scalar multiple, hence } PQ \text{ is parallel to } OB.$$

Worked Example: $OABC$ is a parallelogram with $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = \mathbf{c}$. M is the midpoint of AB and $\overrightarrow{OD} = 2\mathbf{a} + \mathbf{c}$. Prove that O , M and D are collinear.

$$\overrightarrow{AB} = \mathbf{c}, \text{ so } \overrightarrow{OM} = \mathbf{a} + \frac{1}{2}\mathbf{c}$$

$$\overrightarrow{OD} = 2\mathbf{a} + \mathbf{c} = 2(\mathbf{a} + \frac{1}{2}\mathbf{c}) = 2\overrightarrow{OM}$$

$\overrightarrow{OM}$ and $\overrightarrow{OD}$ are parallel and share the point O , so O , M and D are **collinear** (M is the midpoint of OD).

Exam Tip: Proof marks are lost by not finishing the argument. Always end with a sentence: " $\overrightarrow{OD}$ is a multiple of $\overrightarrow{OM}$, so they are parallel; they share point O , so O , M , D are collinear." Showing the vectors are multiples on its own only proves they are parallel.

Summary

Skill	Key fact
Translation	"Translation by $\begin{pmatrix} a \\ b \end{pmatrix}$ " — $(x, y) \rightarrow (x + a, y + b)$
Vector between points	End minus start; $\overrightarrow{BA} = -\overrightarrow{AB}$
Arithmetic	Add/subtract components; scalar multiplies each component
Route on a diagram	$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ when $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$
Parallel Higher	One vector is a scalar multiple of the other

Collinear Higher	Parallel <i>and</i> a common point
Ratio $m : n$ Higher	$\overrightarrow{AP} = \frac{m}{m+n} \overrightarrow{AB}$